

BALAMURUGAN A

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Tirunelveli, Tamil Nadu.

SUMMARY

Mechanical engineering graduate with hands-on experience in CAD design, 3D printing, and prototype development. Skilled in FreeCAD, Python scripting, simulation, and end-to-end prototype development.

INTERNSHIP EXPERIENCE

Pilabz electro mechanical system Pvt. Ltd.

June – December 2025

Mechanical Engineering Intern - Designing

Tenkasi

- Designed and modeled electronic component enclosures in FreeCAD, optimized for 3D printing and practical manufacturability.
- Developed parametric 3D motor components using Python scripting in FreeCAD.
- Executed the complete development of a vacuum forming machine—from mechanical design to a fully functional working prototype.
- Contributed to the mechanical design, assembly, and refinement of a water dispenser prototype.

Bus Depot (Tamil Nadu State Transport Corporation).

July – August 2023

Student Intern – Assembling

Tirunelveli

- Acquired practical experience in the maintenance and operational management of buses.
- Assisted in inspecting vehicle components, monitoring repair activities, and ensuring efficient workflow.

EDUCATION

Government College of Engineering

2021-2025

B.E Mechanical Engineering

Tirunelveli

- CGPA: 7.36

St. Xavier's Higher Secondary School

2019-2021

Higher Secondary Certificate

Tirunelveli

- Percentage: 87.85

Little Flower Matriculation Higher Secondary School

2018

Secondary school certificate

Tirunelveli

- Percentage: 88.2

PROJECT

Vacuum Forming Machine – Working Prototype:

Designed the complete mechanical structure, including the frame, heating chamber, and vacuum chamber, using FreeCAD, with appropriate material selection for prototyping and fabrication. Built and tested a fully functional vacuum forming machine prototype, validating design performance and operational reliability while gaining hands-on experience in thermoforming principles, assembly, and troubleshooting.

SKILLS

- FreeCAD
- Autodesk Fusion 360
- FEM/CFD Simulation
- Python